

EDUCATION

University of Delaware

PhD in Physics and Astronomy, GPA: 3.96/4.0

Newark, DE

2022 – Present

Indian Institute of Science Education Research Bhopal

BS-MS in Physics, CPI: 9.04/10

Bhopal, India

2017 – 2022

PUBLICATIONS (ADS / GOOGLE SCHOLAR)

Articles

(*indicates equal contributions)

- [12] C. R. Bom, A. Mahabal, F. Bianco, P. Darc, B. Fraga, R. Bonito, **S. Chaini**, M. W. Coughlin, S. Dillmann, F. Fontinele Nunes, A. Gomboc, N. Hernitschek, X. Li, F. Z. Majidi, A. I. Malz, A. Melandri, V. Petrecca, S. Piranomonte, M. Rabus, F. Ragosta, O. Razim, M. C. Romão, N. Sarin, A. Sasli, V. A. Srećković, A. Tramuto, V. Vujčić, M. J. Vyas, R. LSST Transients, and Variable Stars Science Collaboration, “Toward decision-aware AI for LSST-scale time-domain astronomy”, *arXiv e-prints*, Jun. 2026. arXiv: [2606.05285](#).
- [11] **S. Chaini**, F. B. Bianco, and A. Mahabal, “Probabilistic Data-Driven Modelling of Astrophysical Transients: The Neural Process Family for Ultrafast and Class-Agnostic Light Curve Reconstruction with NightLANP”, *arXiv e-prints*, May 2026. arXiv: [2605.27527](#).
- [13] N. Earl, **S. Chaini**, K. D. French, J. T. Hinkle, Y. Moon, and M. Shepherd, “NAPTITUDE: A Neural-Process Framework for Rubin Alert Classification”, *arXiv e-prints*, arXiv:2607.19236, arXiv:2607.19236, Jul. 2026. DOI: [10.48550/arXiv.2607.19236](#). arXiv: [2607.19236](#) [[astro-ph.IM](#)].
- [10] E. J. Honaker, J. E. Gizis, C. Aganze, **S. Chaini**, F. B. Bianco, M. Žerjal, E. L. Martín, R. W. Clarke, A. Southwick, H. Petrie, and T. Blask, “Searching for Ultracool Dwarfs in Early LSST Data Products”, *ApJ*, vol. 1004, no. 2, 214, p. 214, Jun. 2026. DOI: [10.3847/1538-4357/ae6ce9](#). arXiv: [2604.08387](#).
- [13] C. L. Ransome, B. M. Subrayan, D. J. Sand, B. Hsu, X. J. Hall, J. Pearson, K. A. Bostroem, J. E. Andrews, J. Vinko, J. C. Wheeler, P. Noel, L. Hu, T. Cabrera, S. Valenti, W. V. Jacobson-Galan, N. Smith, A. V. Filippenko, M. Aghakhanloo, I. Andreoni, M. Andrews, I. Arcavi, R. Baer-Way, E. R. Beasor, E. Berger, F. B. Bianco, P. Blanchard, T. G. Brink, S. Chaini, A. Crawford, Y. Dong, J. Farah, N. Franz, S. Gomez, M. L. Graham, D. Hiramatsu, A. Horti-David, G. Hosseinzadeh, D. A. Howell, P. A. James, S. W. Jha, C. D. Kilpatrick, L. A. Kwok, G. P. Lamb, A. R. Lopes, M. Lundquist, C. E. Martinez-Vasquez, T. Matheson, C. McCully, M. Modjaz, G. S. H. Paek, A. Palmese, A. Patel, J. L. Pledger, A. P. Ravi, J. Rho, K. Sarnecky, M. Shrestha, R. Smith, A. V. Smith Castelli, M. Soraisam, J. Strader, F. Valdes, S. Vasylyev, V. A. Villar, A. K. Vivas, L. Wang, S. D. Wyatt, K. Wynn, and W. Zheng, “First Results from the LSST Shadow Survey: The Restless Luminous Blue Variable AT2017des in the Virgo-Cluster Galaxy, NGC4532”, *arXiv e-prints*, arXiv:2606.23784, arXiv:2606.23784, Jun. 2026. DOI: [10.48550/arXiv.2606.23784](#). arXiv: [2606.23784](#) [[astro-ph.HE](#)].
- [9] **S. Chaini**, F. B. Bianco, and A. Mahabal, “In Search of the Unknown Unknowns: A Multi-Metric Distance Ensemble for Out of Distribution Anomaly Detection in Astronomical Surveys”, *arXiv e-prints*, Oct. 2025. arXiv: [2510.23702](#).
- [8] **S. Chaini**, A. Mahabal, A. Kembhavi, and F. B. Bianco, “Light curve classification with DistClassiPy: A new distance-based classifier”, *Astronomy and Computing*, vol. 48, 100850, p. 100 850, Jul. 2024. DOI: [10.1016/j.ascom.2024.100850](#). arXiv: [2403.12120](#).

- [7] B. Domínguez*, **S. Chaini***, K. Mrakovčić*, B. Sallee, and Ž. Ivezić, “Validation of the Bond et al. (2010) SDSS-derived kinematic models for the Milky Way’s disk and halo stars with Gaia Data Release 3 proper motion and radial velocity data”, *A&A*, vol. 688, A132, A132, Aug. 2024. DOI: [10.1051/0004-6361/202449653](https://doi.org/10.1051/0004-6361/202449653). arXiv: [2406.03541](https://arxiv.org/abs/2406.03541).
- [6] **S. Chaini**, A. Bagul, A. Deshpande, R. Gondkar, K. Sharma, M. Vivek, and A. Kembhavi, “Photometric identification of compact galaxies, stars, and quasars using multiple neural networks”, *MNRAS*, vol. 518, no. 2, pp. 3123–3136, Jan. 2023. DOI: [10.1093/mnras/stac3336](https://doi.org/10.1093/mnras/stac3336). arXiv: [2211.08388](https://arxiv.org/abs/2211.08388).
- [5] **S. Chaini** and S. S. Kumar, “Astronomical Classification of Light Curves with an Ensemble of Gated Recurrent Units”, *arXiv e-prints*, Jun. 2020. arXiv: [2006.12333](https://arxiv.org/abs/2006.12333).

Software

- [3] **S. Chaini**, F. B. Bianco, and A. Mahabal, “*NightLANP*”, GitHub repository, Jun. 2026. [Online]. Available: <https://github.com/sidchaini/NightLANP/>.
- [2] **S. Chaini**, F. B. Bianco, and A. Mahabal, “*keras-neural-processes*”, GitHub repository, Nov. 2024. [Online]. Available: <https://github.com/sidchaini/keras-neural-processes/>.
- [1] **S. Chaini**, A. Mahabal, A. Kembhavi, and F. B. Bianco, “*DistClassiPy*”, Astrophysics Source Code Library, record ascl:2403.002, Mar. 2024. [Online]. Available: <https://github.com/sidchaini/DistClassiPy/>.

HONORS AND AWARDS

- [Future Investigators in NASA Earth and Space Science and Technology \(FINESST\)](#), NASA 2025 – 2027
- [Doctoral Fellowship for Excellence](#), University of Delaware 2025 – 2026
- [Data Science Institute Fellow](#), University of Delaware 2023 – Present
- Vatican Observatory Scholarship for attending the [18th Vatican Observatory Summer School](#) 2023
- [INSPIRE](#) Scholarship for Higher Education, DST, Government of India 2017–2022

ACADEMIC SERVICE

- **Elected Member** of the LSST ISSC Executive Council Jun. 2026 – Present
Responsible for oversight of working groups, committee appointments, and collaboration policies
Informatics and Statistics Science Collaboration
- **Co-Chair** of the LSST TVS Anomalies Subgroup Sep. 2024 – Present
Involved in organizing meetings for the Anomalies subgroup of the Rubin LSST
Transients and Variables Science Collaboration
- **Elected Head** of the Student Research Group at IISER Bhopal Astronomy Club Aug. 2020 – Aug. 2021
In charge of group activities and data analysis - image reduction and CCD photometry

TALKS AND PANELS

- Rubin Community Workshop 2026 - Contributed talk Jul. 2026
Fast, Probabilistic Modelling for the LSST Alert Stream - Interpolating Sparse Multiband Light Curves with Neural Processes
- The Extragalactic Transient Universe - Contributed talk Jul. 2026
Probabilistic Modelling of Transient Light Curves with Neural Processes
- European Astronomical Society Annual Meeting - Contributed talk Jun. 2026
Using neural processes to model sparse light curves for LSST-like transients

- American Physical Society Global Physics Summit - Contributed talk Mar. 2026
Tackling Sparse, Multiwavelength Light Curves in Time-Domain Astronomy with Neural Processes
- TDABench - Contributed talk Mar. 2026
The Need for a Light Curve Interpolation and Forecasting Benchmark Dataset
- Quasar Bazaar Hackweek - **Invited talk** (remote) Feb. 2026
Distance Multi-Metric Anomaly Detection (DiMMAD)
- NCRA Astronomy Seminar - **Invited talk** Jan. 2026
Transient hunting in the era of large-sky surveys with Neural Processes
- LSST@Europe 7 - **Contributed talk** Sep. 2025
Tackling LSST's sparse multiband light curves with Neural Processes
- Rubin Community Workshop 2025 - **Contributed talk** Jul. 2025
Hunting for Anomalies with Distance Metrics for LSST
- Shaping the Future of Time-Domain Astronomy with LSST - Contributed talk May 2025
Anomalies in the time-domain: Machine learning with distance metrics
- AI/ML Applications in Astronomy and Astrophysics - Panel Discussion Jan. 2025
Pitfalls of using ChatGPT
- AI/ML Applications in Astronomy and Astrophysics - **Hands-On Tutorial** Jan. 2025
Leveraging Distance Metrics for Better Machine Learning
- LSST TVS - **Colloquium** Oct. 2024
Distance Metrics for Machine Learning in Time-Domain Astronomy with DistClassiPy
- LSST@Europe 6 - **Contributed talk** Sep. 2024
Leveraging Distance Metrics for Machine Learning in Astronomy: DistClassiPy
- Rubin Community Workshop 2024 - **Contributed talk** Jul. 2024
DistClassiPy: Distance-Based Techniques for Classification and Anomaly Detection in Light Curves

POSTERS

- EAS 2026: What physics can we find in the data-driven manifolds of supernovae spectra? Jul. 2026
- IAU Symposium 406: Future landscape of astrophysical transients - **Poster** May. 2026
- NeurIPS 2025 (*Machine Learning for Physical Sciences*) - **Poster** Dec. 2025
- Data Science & DARWIN Computing Symposium - Poster Apr. 2025
- Rare Gems in Big Data - **Poster** May 2024
- 2023 Annual Meeting of the APS Mid-Atlantic Section - **Poster** Nov. 2023
- 2023 Delaware Data Science Symposium - Poster Sep. 2023
- 40th Meeting of the Astronomical Society of India - **Poster** Mar. 2022
- 2021 Institut Astrophysique de Paris Colloquium - **Poster** Oct. 2021

MENTORING

- Co-mentoring Lincoln Undergrads Abdourahim Sylla, Agustina Lorda, Adeoluwa Afolabi May. 2026 – Aug. 2026
- Co-mentored UD Undergraduate NASA Space Grant Fellow, Ibrahim Wilson Jul. 2023 – Aug. 2024

TEACHING EXPERIENCE

- **Online Instructor**, LSST TVSSC Jul. 2024 - Present
Intermediate Python for Astronomical Software Development
- **Teaching assistant**, University of Delaware Sep. 2022 - Aug. 2024
PHYS 207, PHYS 208, PHYS 227, SCEN 101
- **Teaching assistant**, IISER Bhopal Jan. 2019 – May 2019
ECS 102

OUTREACH

- **Rubin First Look** at UD
 - Involved in organizing telescope viewing nights for the IISER Bhopal community
 - Involved in organizing programming tutorials on computational astronomy for IISER Bhopal students
- **Astronomy Club** at IISER Bhopal
 - Involved in organizing telescope viewing nights for the IISER Bhopal community
 - Involved in organizing programming tutorials on computational astronomy for IISER Bhopal students
- **Outreach Talks** at local tutoring centres in Thane, India
 - Gave science talks to students in 8th-10th grade, sharing insights and inspiring interest in science and astrophysics

MEDIA COVERAGE

- UDaily, “*NASA’s next generation of scientists*” Oct. 2025
- UDaily, “*Opening a new window into the Universe*” Jun. 2025
- Franciscan Media, “*Vatican hosts astronomers of tomorrow for summer school*” [Also on [YouTube](#)] Jun. 2023

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Wolfram Language, HTML, SQL, $\text{\LaTeX}$, Assembly Language, Bash

Libraries: Astropy, NumPy, Keras, TensorFlow, PyTorch, pandas, scikit-learn, Selenium, matplotlib

Software: Mathematica, SAOImage DS9, Aperture Photometry Tool

Developer Tools: Git, Jupyter, VS Code